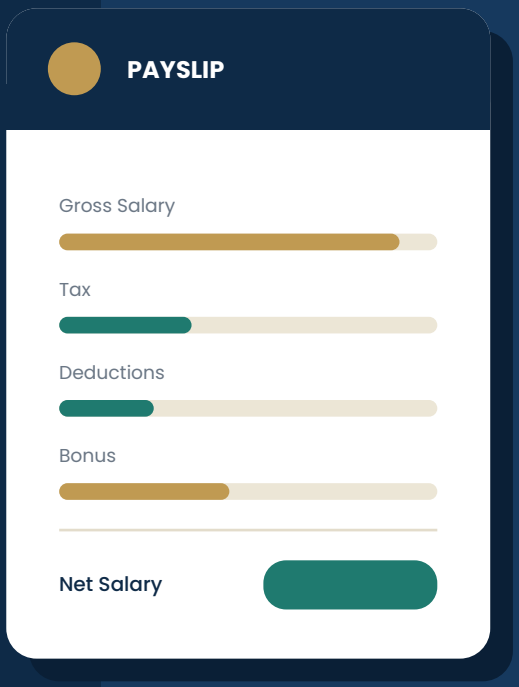
MILESTONE #1

Payroll Management System

UML Class Diagram and Class Descriptions



MILESTONE #1

Payroll Management System

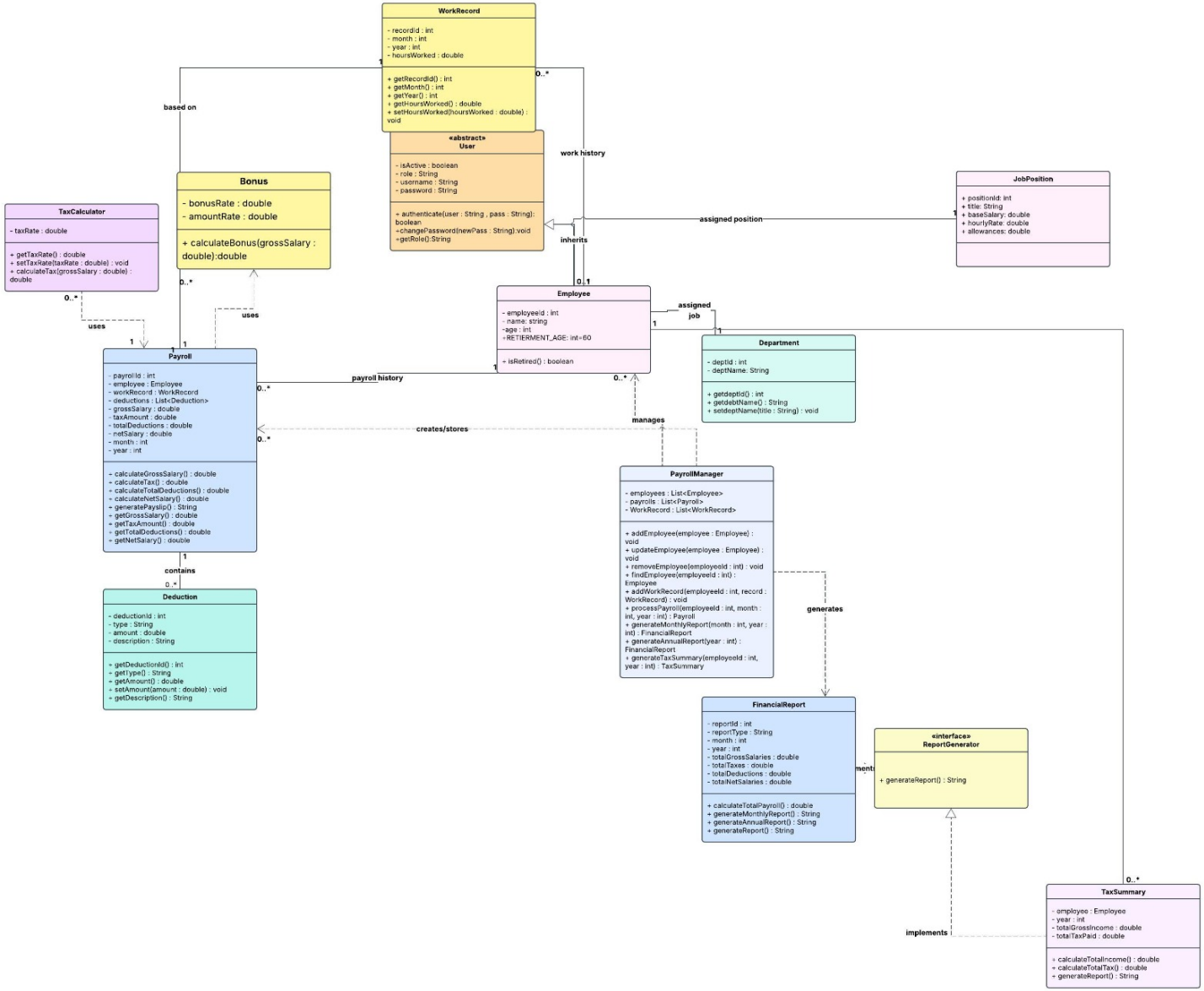
UML Class Diagram and Class Descriptions

CLASSES AT A GLANCE

#	Class	Type	Group
01	**User**	Abstract Class	People & Access
02	**Employee**	Class	People & Access
03	**Department**	Class	People & Access
04	**JobPosition**	Class	People & Access
05	**WorkRecord**	Class	Work & Compensation
06	**Payroll**	Class	Work & Compensation
07	**Bonus**	Class	Work & Compensation
08	**TaxCalculator**	Class	Work & Compensation
09	**Deduction**	Class	Work & Compensation
10	**PayrollManager**	Class	Coordination & Reporting
11	**TaxSummary**	Class	Coordination & Reporting
12	**FinancialReport**	Class	Coordination & Reporting
13	**ReportGenerator**	Interface	Coordination & Reporting

UML CLASS DIAGRAM

Payroll Management System — Class Diagram



MILESTONE #1

Class Descriptions

PEOPLE & ACCESS

01 User

ABSTRACT CLASS

Defines shared account information and authentication-related behavior for system users, including username, password, role, and account status. Employee inherits these common properties.

02 Employee

CLASS

Represents a company employee, storing personal details, age, and retirement information. An employee inherits from User, is assigned a job position, belongs to a department, and is associated with work records, payroll history, and tax summaries.

03 Department

CLASS

Stores department information, including its identifier and name. Multiple employees may belong to the same department, which groups employees organizationally.

04 JobPosition

CLASS

Defines a job position, including its identifier, title, base salary, hourly rate, and allowances. Each employee is assigned a position, which supplies the pay-related values that can be used when calculating pay.

WORK & COMPENSATION

05 WorkRecord

CLASS

Records an employee's working hours for a specified month and year. It provides the work data used for payroll calculation.

06 Payroll

CLASS

Represents the salary calculation and payslip for one employee during a specific month. It calculates gross salary, tax, deductions, and net salary using the associated work record and supporting calculation components.

07 Bonus

CLASS

Models additional compensation using a bonus rate or amount. It calculates a bonus that can be included in an employee's payroll.

08 TaxCalculator

CLASS

Encapsulates the tax rate and calculates tax on a supplied taxable salary amount, keeping tax calculation separate from the main payroll workflow.

09 Deduction

CLASS

Represents an individual non-tax salary deduction, such as insurance, a loan repayment, or a penalty. A payroll may contain multiple deductions.

COORDINATION & REPORTING

10 PayrollManager

CLASS

Coordinates payroll-related operations. It manages collections of employees, work records, and payroll entries, processes payroll requests, and creates financial reports and employee tax summaries.

11 TaxSummary

CLASS

Summarizes an employee's annual gross income and tax paid. It implements ReportGenerator to produce a readable individual tax report.

12 FinancialReport

CLASS

Summarizes company-wide payroll figures, including gross salaries, taxes, deductions, and net salaries. It generates monthly and annual financial reports and implements ReportGenerator.

13 ReportGenerator

INTERFACE

Defines the generateReport() contract shared by report-producing classes. FinancialReport and TaxSummary implement this interface, supporting polymorphic report generation.